UNIVERSITY OF RWANDA
COLLEGE OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTER AND SOFTWARE ENGINEERING

Mobile Application System and Design

Flutter Lab 3b

Multi-Screen Travel App UI

1. Introduction

This App is Multi-Screen Travel App UI built using Flutter and Dart. The app was developed as part of Flutter Lab 3b, focusing on advanced UI design, widget composition, and navigation between multiple screens.

The app simulates a travel booking application with three main screens: Home Screen, Detail Screen, and Booking Screen. All data is hard-coded inside Dart files with no API or database required.

2. Lab Objective

The main objectives of this lab are:

- Build a multi-screen Travel App UI with hard-coded data
- Practice complex layouts and widget composition in Flutter
- Implement navigation between multiple screens
- Use at least 18 different Flutter widgets
- Create custom reusable widget classes
- Apply gradients, shadows, and consistent color themes

3. App Structure and Folder Organization

The project follows a clean and organized folder structure:

Folder/File	Description
lib/main.dart	Entry point of the application
lib/screens/home_screen.dart	Home Screen with destination grid
lib/screens/detail_screen.dart	Detail Screen for each destination
lib/screens/booking_screen.dart	Booking Screen with confirmation
lib/widgets/destination_card.dart	Reusable destination card widget
lib/widgets/category_chip.dart	Reusable category filter chip widget
lib/data/travel_data.dart	Hard-coded destination data file
assets/images/	Folder containing all destination images

4. Hard-Coded Data

All data is stored in lib/data/travel_data.dart. The file contains a Destination class and a list of 5 hard-coded destinations:

Destination	Country	Category	Price	Rating

Bali	Indonesia	Beach	$200	4.8
Paris	France	City	$500	4.9
Dubai	UAE	City	$800	4.7
Tokyo	Japan	Culture	$600	4.9
Rome	Italy	Culture	$450	4.8

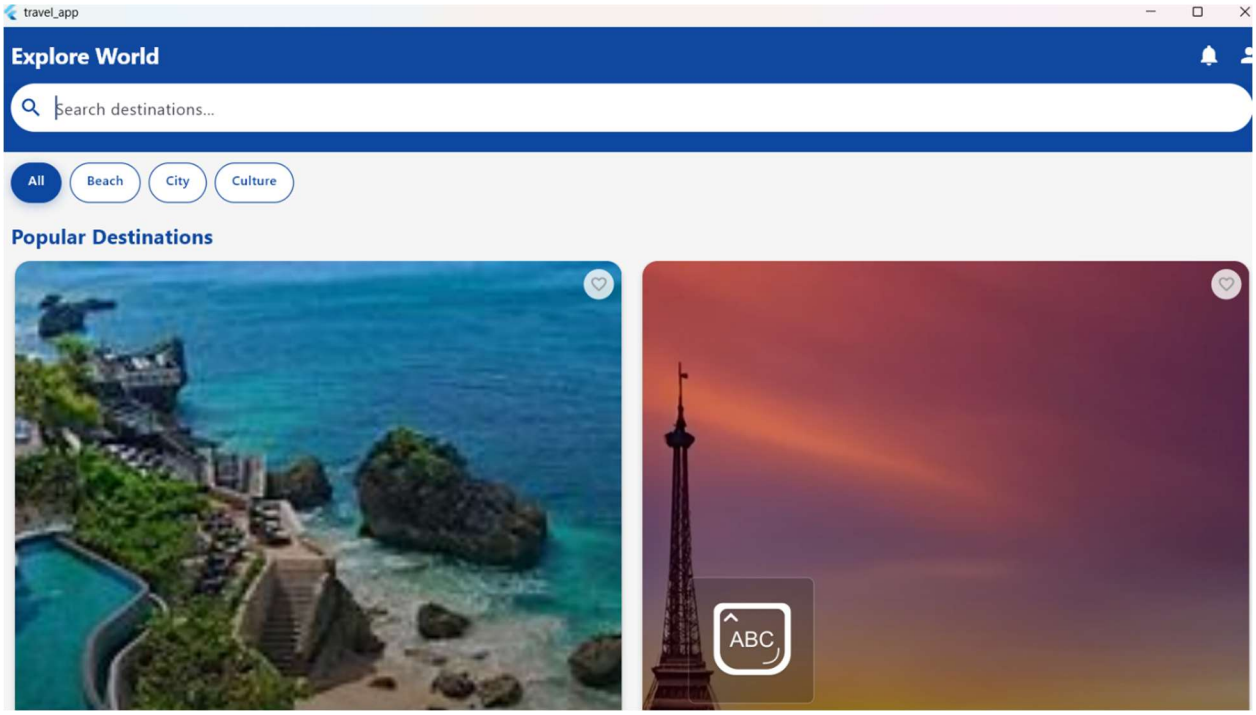
5. Required Screens

5.1 Home Screen

The Home Screen serves as the main dashboard of the application. It includes:

- AppBar with title 'Explore World' and notification and profile icons
- Search bar that filters destinations by name, country, or category
- Horizontal category scrolling with All, Beach, City, and Culture options
- Grid of destination cards using GridView.builder
- Each card shows an image, destination name, country, rating, price, and favorite icon

Layout Choice: A GridView with 2 columns was chosen to display destinations efficiently. Stack and Positioned widgets were used to overlay text and favorite buttons on top of images. A gradient overlay was added for better text readability.

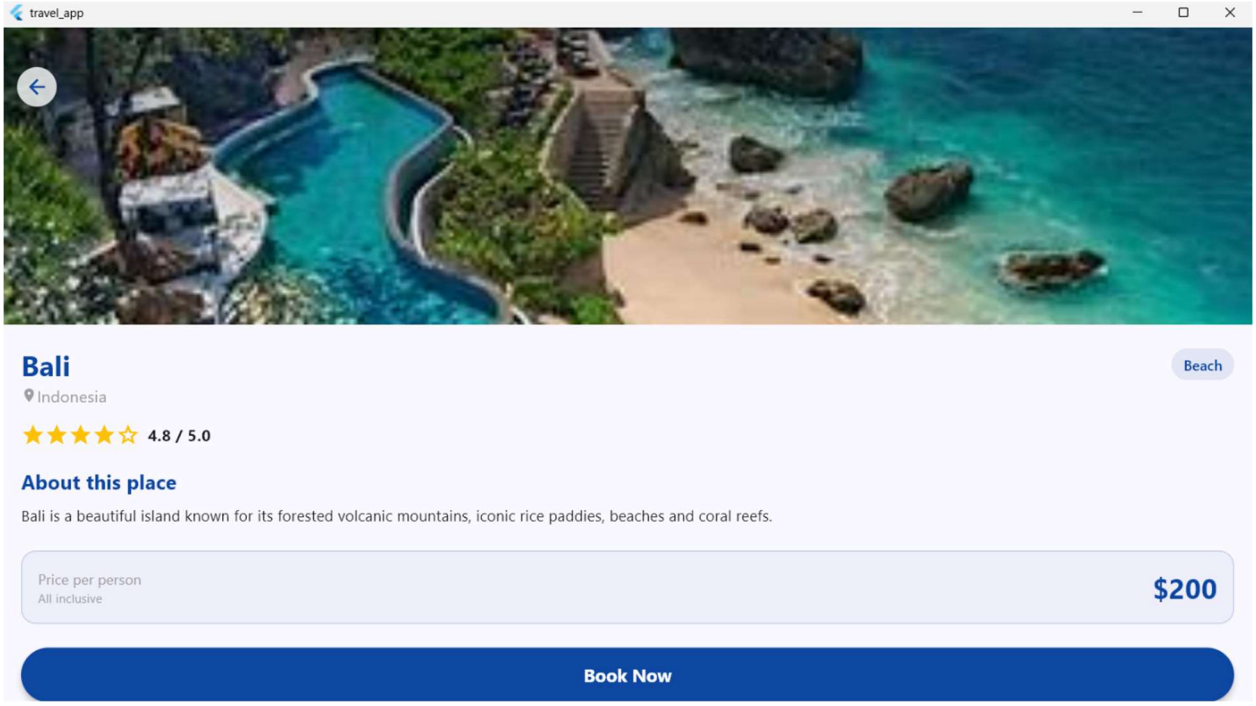


5.2 Detail Screen

The Detail Screen displays comprehensive information about a selected destination:

- Large full-width header image at the top
- Back navigation button overlaid on the image
- Destination name, country, and category badge
- Star rating display using dynamic icon generation
- Scrollable description text using SingleChildScrollView
- Price section with 'All inclusive' label
- Styled 'Book Now' button that navigates to Booking Screen

Layout Choice: SingleChildScrollView wraps a Column to allow scrolling on smaller screens. Stack was used to overlay the back button on the header image. Custom reusable widgets were created for the rating and price sections.

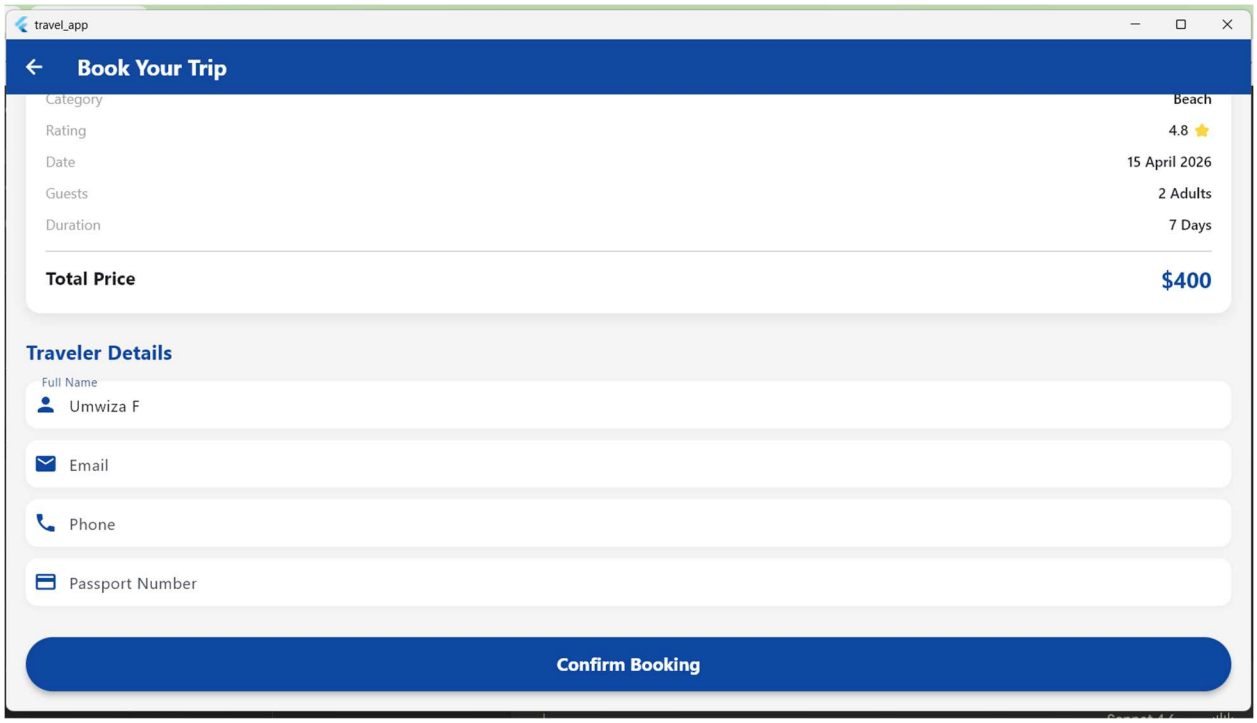


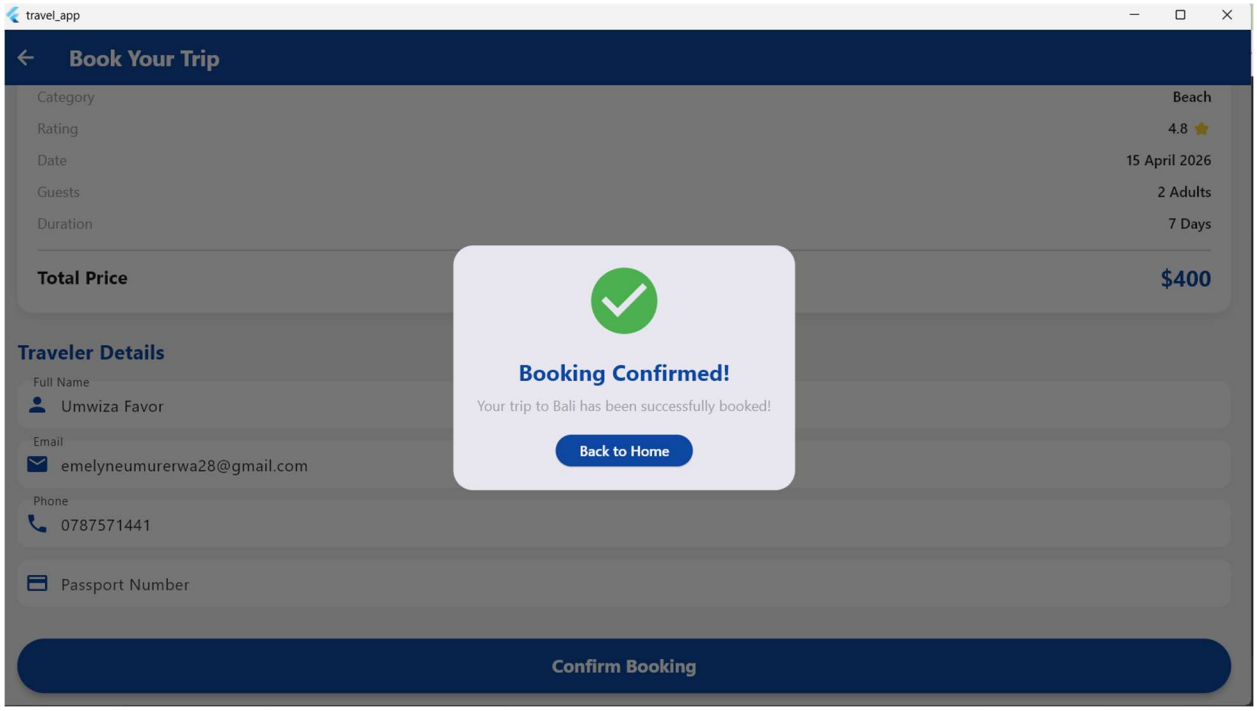
5.3 Booking Screen

The Booking Screen displays a booking summary and form:

- Hard-coded booking summary card with destination details
- Total price calculation (price x 2 guests)
- Form-style layout with text fields for traveler details
- Confirm Booking button
- Success dialog with green checkmark icon after confirmation
- 'Back to Home' button in dialog that returns to Home Screen

Layout Choice: A Card container with shadow was used for the booking summary to create visual separation. Form fields use a consistent style with prefix icons and rounded borders.





6. Navigation Flow

The app implements navigation using Navigator.push and Navigator.pop as required:

From	To	Trigger	Method
Home Screen	Detail Screen	Tap destination card	Navigator.push
Detail Screen	Booking Screen	Tap Book Now button	Navigator.push
Booking Screen	Detail Screen	Tap Back button	Navigator.pop
Booking Screen	Home Screen	Tap Back to Home in dialog	Navigator.popUntil
Detail Screen	Home Screen	Tap Back button	Navigator.pop

7. List of Flutter Widgets Used

These are more than 18 different flutter widgets which are used :

#	Widget	Where Used
1	MaterialApp	main.dart - App entry point
2	Scaffold	All screens - page structure
3	AppBar	Home Screen and Booking Screen
4	Container	Search bar, cards, price section
5	Column	Layout in all screens
6	Row	Navigation bar, rating, price row
7	Stack	Image with overlay text and buttons
8	Positioned	Back button and card text overlay
9	GridView.builder	Home Screen destination grid
10	ListView.builder	Category horizontal scroll
11	Card	Destination cards and booking summary
12	GestureDetector	Card tap and favorite button tap
13	Image.asset	Destination images
14	Text	All text content
15	Icon	Favorite, location, star, back icons
16	IconButton	AppBar notification and profile icons
17	ElevatedButton	Book Now and Confirm Booking buttons
18	TextField	Search bar and booking form fields
19	SingleChildScrollView	Detail Screen and Booking Screen
20	ClipRRect	Rounded image corners on cards

21	SizedBox	Spacing between widgets
22	Padding	Content padding
23	Expanded	Fill available space in layouts
24	AlertDialog	Booking confirmation popup
25	Divider	Separator in booking summary

8. UI Design Decisions

8.1 Color Theme

A consistent blue color theme was applied throughout the app:

- Primary color: #0D47A1 (Dark Blue) - used for AppBar, buttons, and headings
- Background: #F5F5F5 (Light Grey) - used for screen backgrounds
- Accent: Green for prices, Amber for star ratings

8.2 Gradients

Linear gradients were used in two places:

- Destination cards: A gradient from transparent to black overlays the bottom of images to make text readable
- Detail Screen header: A gradient from semi-transparent black to transparent at the top of the header image

8.3 Shadows

Box shadows were applied to:

- Destination cards using Card elevation property
- Category chips when selected
- Booking summary container
- Form fields in the Booking Screen

8.4 Custom Reusable Widgets

Two custom reusable widget classes were created:

- DestinationCard - displays a destination with image, name, rating, price, and favorite button
- CategoryChip - displays a category filter button with selected/unselected states

9. App Screenshots

Note: Please attach the following screenshots to this report:

Screenshot	Description
screenshot_home.png	Home Screen showing all 5 destination cards in a grid
screenshot_detail.png	Detail Screen showing Bali destination information
screenshot_booking.png	Booking Screen showing summary and form fields
screenshot_confirmation.png	Confirmation dialog after booking is confirmed

-

10. Conclusion

This lab successfully demonstrates the implementation of a multi-screen Travel App UI using Flutter and Dart.

- Three fully functional screens with smooth navigation
- Hard-coded data stored in a dedicated Dart file
- More than 18 different Flutter widgets used
- Custom reusable widget classes created
- Gradients and shadows applied for visual appeal
- Consistent color theme throughout the app
- Images loaded from local assets folder
- Navigator.push and Navigator.pop for navigation